Problem Set 1

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Advanced Methods in Applied Statistics

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Problem 1

The histograms of the Adjusted Defense Efficiency "DRtg" for all the teams in the 5 conferences (ACC, SEC, B10, BSky, and A10) are shown in fig. 1. In order to distinguish between different conferences different hatches are used for each conference as evident by the legend.

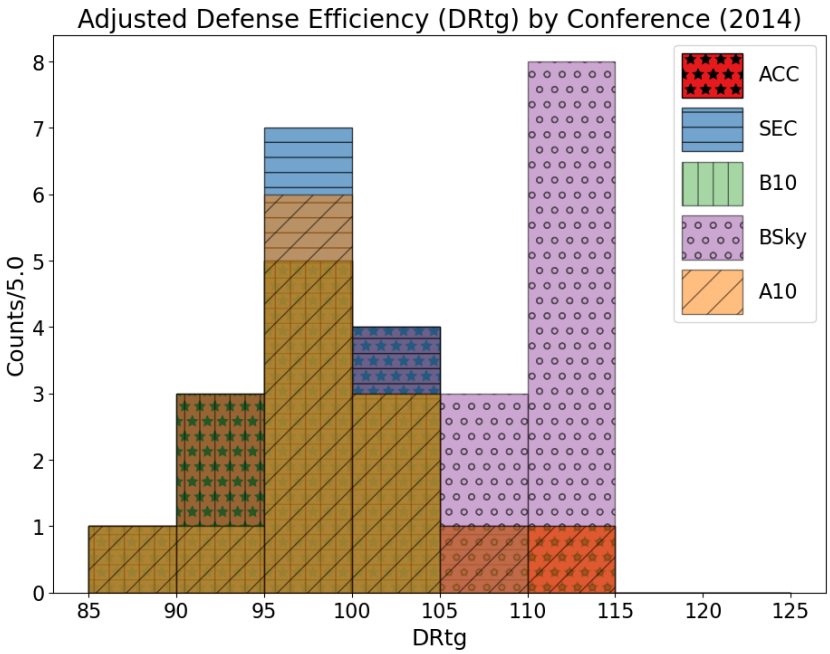

Figure 1: Histogram showing the Adjusted Defense Efficiency "DRtg" for all the teams in the 5 conferences (ACC, SEC, B10, BSky, and A10)

Problem 2

Some teams didn't feature in both the 2009 and 2014 season and were removed from the data set accordingly. Some teams also changed conferences so that they weren't in the 5 conferences (ACC, SEC, B10, BSky, and A10) in both 2009 and 2014. Only teams that were in the same of one of the 5 conferences in both 2009 and 2014 were included in the data points shown in fig. 2. Teams that changed conferences, but were still in both seasons were included when calculating the mean and median of differences in "ORtg" for teams not in the 5 conferences (ACC, SEC, B10, BSky, and A10). The means and medians of these other teams as well as the ones from each of the conferences are listed in the table table 1.

Conference	Mean	Median
ACC	-0.625	-0.05
SEC	2.283	1.65
B10	2.673	4.3
BSky	2.322	3.2
A10	4.336	4.9
Other	2.382	1.3

Table 1: Medians and means of the difference in "ORtg" between 2009 and 2014

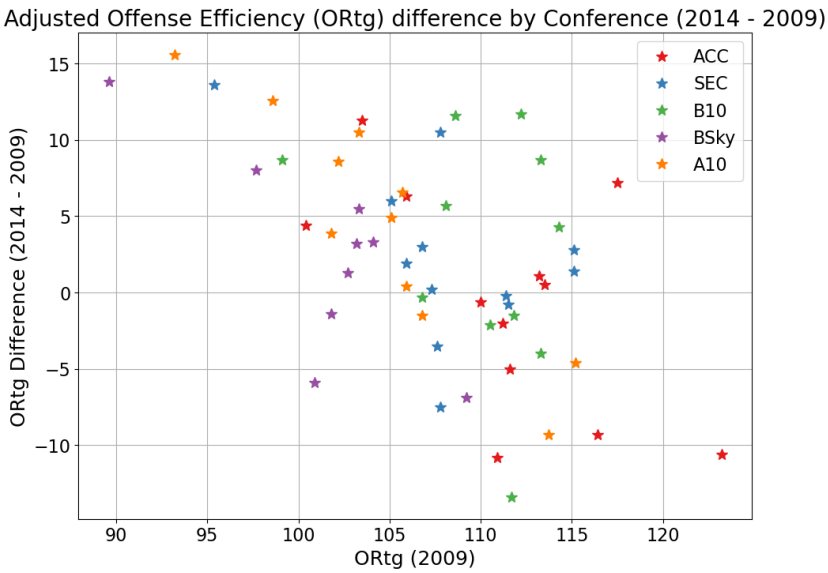

Figure 2: Differences in "ORtg" for all the teams in the 5 conferences (ACC, SEC, B10, BSky, and A10) between 2009 and 2014.

Problem 3

Once again hatches are used to distinguish between histograms from different conferences shown in fig. 3. Likewise for the difference in "ORtg" only teams that were in in the same of one of the 6 conferences in both 2009 and 2014 were included in the data points shown in fig. 4. The mean and median for BE and the other teams (now also excluding teams from BE) are listed in table 2.

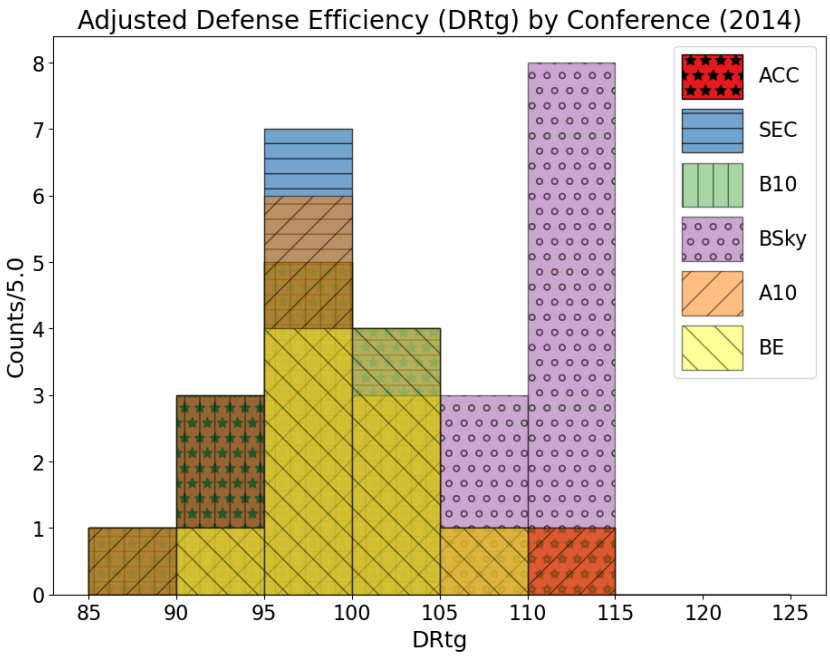

Figure 3: Histogram showing the Adjusted Defense Efficiency "DRtg" for all the teams in the 6 conferences (ACC, SEC, B10, BSky, A10, and BE)

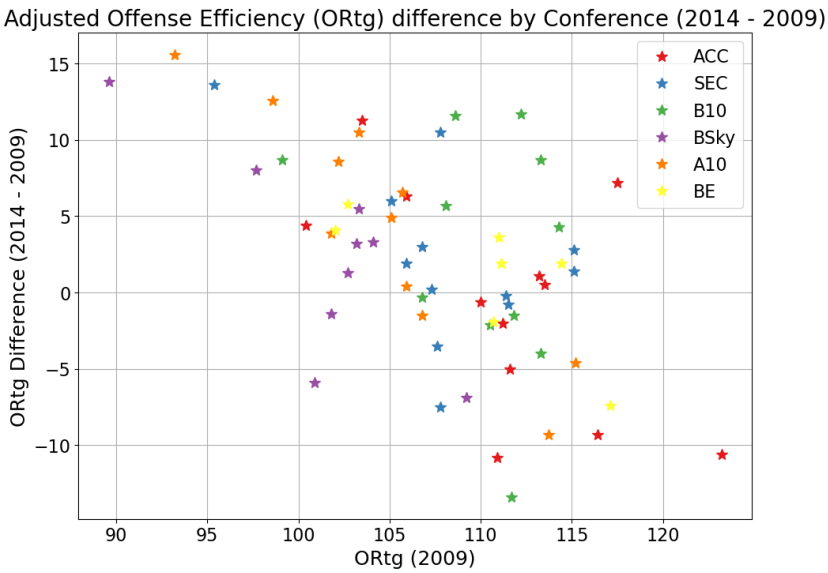

Figure 4: Differences in "ORtg" for all the teams in the 6 conferences (ACC, SEC, B10, BSky, A10, and BE) between 2009 and 2014.

Conference	Mean	Median
BE	1.143	1.9
Other	2.413	1.3

Table 2: Medians and means of the difference in "ORtg" between 2009 and 2014

Extra credit

There are 3614 unique authors in the author list.
Since there is a even number of authors there are two mid-point authors. These are "K. Link" and "S. D. Linker".